Machine Learning and Deep Learning Approaches for Loan Approval Prediction: A Comparative Study

Riyadh Hossain, Mahmudul Hasan Tanvir, Walid Ahmed

Dept of Computer science & Engineering

East West University,

Abstract— Timely and accurate assessment of loan applicants is a critical task for financial institutions seeking to minimize default risk while extending credit access. This paper presents a comparative study of three supervised learning models—a multilayer perceptron (referred to in the source implementation as a CNN-style dense network), a deeper artificial neural network (ANN) with early stopping, and a Random Forest ensemble classifier—for binary loan-approval prediction. Using a public dataset of 45,000 loan applications described by thirteen demographic, financial, and credit-history features, categorical variables were label-encoded and numerical features were standardized prior to an 80/20 stratified train-test split. All three models were trained and evaluated under identical data conditions. The Random Forest classifier achieved the highest test accuracy of 92.77%, outperforming the multilayer perceptron (91.76%) and the deeper ANN (91.44%). Feature-importance analysis further revealed that prior loan default history, loan-to-income ratio, and interest rate are the strongest predictors of loan status. These findings suggest that ensemble tree-based methods remain highly competitive with, and in this case superior to, deep neural architectures for structured tabular credit-risk data. To strengthen transparency and support responsible use of automated lending, the study is extended with an Explainable AI (XAI) framework based on SHAP (SHapley Additive exPlanations). The proposed XAI analysis provides both global feature-attribution summaries and local explanations for individual loan decisions, while the existing Random Forest impurity-based importance is retained as a baseline. The SHAP results should be generated from the exact trained model and held-out test set before final journal submission; no uncomputed SHAP values are reported here.

Index Terms— loan approval prediction, credit risk, machine learning, deep learning, artificial neural network, random forest, feature importance, classification, explainable artificial intelligence, SHAP, interpretability

I. INTRODUCTION

Access to credit is a cornerstone of individual and business financial activity, yet lending decisions carry inherent risk for the institutions that extend it. An inaccurate approval decision may result either in default losses, when a high-risk applicant is approved, or in lost revenue and reduced financial inclusion, when a creditworthy applicant is rejected. Traditional credit-scoring approaches rely on manually engineered rules and logistic scoring models; however, the growing availability of large, feature-rich applicant datasets has motivated the adoption of machine learning (ML) and deep learning (DL) techniques that can capture complex, non-linear relationships between applicant attributes and repayment behavior.

This paper investigates and compares the performance of three classification approaches on a large-scale loan applicant dataset: (i) a multilayer perceptron with dropout regularization, (ii) a deeper artificial neural network trained with early stopping, and (iii) a Random Forest ensemble classifier. The objective is to determine which modeling paradigm—deep neural networks or tree-based ensembles—provides the most accurate and interpretable predictions for structured, tabular credit data, and to identify the applicant attributes most predictive of loan outcome.

The remainder of this paper is organized as follows. Section II describes the dataset and exploratory characteristics. Section III details the data preprocessing pipeline. Section IV describes the methodology and model architectures. Section V presents the experimental setup. Section VI reports and discusses the results. Section VII concludes the paper and outlines directions for future work.

II. RELATED WORK

Machine learning has become increasingly popular in loan approval and credit-risk prediction, largely because it can make lending decisions more accurate and efficient than conventional approaches. Traditional credit-scoring methods typically rely on statistical models and fixed rule sets, whereas machine-learning approaches are better suited to capturing the nonlinear relationships that often exist between financial and applicant-related attributes. Random Forest, in particular, has become a popular choice for classification tasks thanks to its ensemble structure and its ability to handle complex interactions between features. Artificial neural networks are also commonly used in financial prediction tasks, given their capacity to learn nonlinear patterns in structured data.

A number of prior studies have shown that machine-learning and deep-learning methods hold real promise for credit-risk

assessment. And with that, how well a model performs can depend heavily on factors such as the dataset used, the nature of its features, the preprocessing steps applied, and how the model itself is configured. This makes it worthwhile to compare different approaches under the same experimental conditions, since doing so helps clarify which method is actually better suited to a given loan-approval dataset.

In this study, we evaluate three supervised learning approaches a Multilayer Perceptron, a deeper Artificial Neural Network trained with early stopping, and a Random Forest classifier all using the same dataset, preprocessing pipeline, and stratified train-test split. We also examine Random Forest's feature importance scores to identify which applicant attributes contribute most to the model's predictions.

III. DATASET DESCRIPTION

The dataset used in this study comprises 45,000 loan applicant records, each described by thirteen input features and a binary target variable, loan_status (1 = approved, 0 = rejected). The dataset contains no missing values across any of the fourteen columns. Table I summarizes the input features grouped by category, and Fig. 1 shows the distribution of the target class.

The features span four broad categories: (1) demographic attributes (age, gender, education level, employment experience), (2) financial attributes (annual income, home-ownership status, loan amount, loan intent), (3) loan-specific attributes (interest rate, loan-to-income percentage), and (4) credit-history attributes (credit history length, credit score, and prior default record). The target variable is imbalanced, with approximately 77.8% of applicants labeled as rejected (0) and 22.2% labeled as approved (1), reflecting the class distribution typical of real-world credit-risk datasets.

Table I. Summary of Dataset Features

Feature	Type	Description
person_age	Numeric	Applicant age (years)
person_gender	Categorical	Male / Female
person_education	Categorical	High School–Doctorate
person_income	Numeric	Annual income (USD)
person_emp_exp	Numeric	Employment experience (yrs)
person_home_ownership	Categorical	Rent/Own/Mortgage/Other
loan_amnt	Numeric	Requested loan amount
loan_intent	Categorical	Purpose of loan
loan_int_rate	Numeric	Interest rate (%)
loan_percent_income	Numeric	Loan-to-income ratio
cb_person_cred_hist_length	Numeric	Credit history length (yrs)
credit_score	Numeric	Credit bureau score
previous_loan_defaults_on_file	Categorical	Yes / No

IV. DATA PREPROCESSING

Prior to model training, all categorical variables—person_gender, person_education, person_home_ownership, loan_intent, and previous_loan_defaults_on_file—were transformed into numeric representations using scikit-learn's Label Encoder. An exhaustive check confirmed zero missing values across all 45,000 records, so no imputation was required.

The processed dataset was split into a feature matrix X (13 columns) and a target vector y (loan_status). An 80/20 train-test split was performed using stratified sampling on the target variable to preserve class proportions, yielding 36,000 training samples and 9,000 testing samples. For the neural network models, StandardScaler was applied to normalize all feature values to zero mean and unit variance, using statistics computed only on the training partition and subsequently applied to the test partition to avoid data leakage.

V. METHODOLOGY

A. Multilayer Perceptron (MLP)

The first neural architecture is a fully connected feedforward network with an input layer sized to the thirteen input features, followed by dense layers of 64, 32, and 16 ReLU-activated units interleaved with dropout layers (rates 0.3 and 0.2) for regularization, and a single sigmoid output unit for binary classification. The network was compiled with the Adam optimizer and binary cross-entropy loss, and trained for 100 epochs with a batch size of 32 and a 20% validation split.

B. Deep Artificial Neural Network (ANN)

The second architecture follows a similar layer structure (64–32–16 ReLU units with dropout of 0.3 and 0.2) but incorporates an early-stopping callback that monitors validation loss with a patience of five epochs and restores the best-performing weights. This mechanism halts training once validation performance stops improving, mitigating overfitting and reducing unnecessary computation.

C. Random Forest Classifier

The third model is a Random Forest ensemble composed of 100 decision trees (n_estimators = 1000) trained using the Gini impurity criterion with unrestricted maximum depth. As an ensemble of bootstrap-aggregated decision trees, the Random Forest additionally provides an interpretable feature-importance ranking derived from the mean decrease in impurity across all trees.

VI. EXPERIMENTAL SETUP

All experiments were implemented in Python using TensorFlow/Keras for the neural network models and scikit-learn for preprocessing and the Random Forest classifier. A fixed random seed (random_state = 42) was used for the train-test split and the Random Forest to ensure reproducibility. Model performance was evaluated using

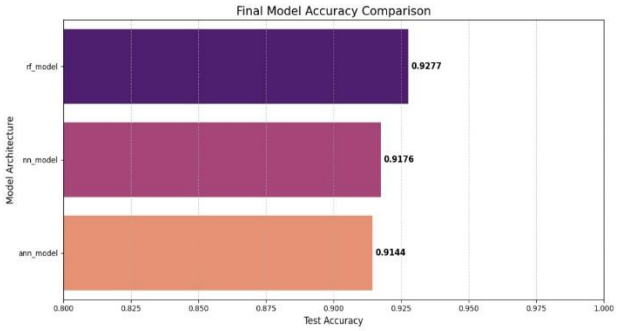
classification accuracy on the held-out 9,000-sample test set, computed as the proportion of correctly predicted loan-status labels. Model performance was assessed using accuracy, precision, recall, and F1-score on a held-out test set of 9,000 samples, complemented by confusion matrices to examine each classifier's correct and incorrect predictions across both loan-status classes. This combination of metrics was chosen in light of the dataset's class imbalance — approximately 77.8% of applications were labeled rejected, against 22.2% approved — since accuracy alone can obscure a model's true performance under such conditions.

VII. RESULTS AND DISCUSSION

Table II summarizes the final test-set accuracy achieved by each of the three models, and Fig. 2 presents the corresponding comparison chart.

Table II. Final Model Test Accuracy Comparison

Model	Test Accuracy
Random Forest	0.9277
Multilayer Perceptron	0.9176
Deep ANN (early stopping)	0.9144



Model	Accuracy	Precision (1)	Recall (1)	F1-score (1)
Multilayer Perceptron	91.76%	0.86	0.75	0.80
Deep ANN	91.44%	0.86	0.73	0.79
Random Forest	**92.77%**	**0.89**	**0.77**	**0.83**

Fig. 2. Final test accuracy comparison across the three classification models.

The Random Forest classifier achieved the highest test accuracy at 92.77%, outperforming both neural network models by approximately one to two percentage points. This result suggests that, for this moderately sized tabular dataset with a mix of categorical and numeric features, an ensemble of decision trees captures the relevant feature interactions at least as effectively as a fully connected neural network, while requiring substantially less computational overhead and no feature scaling.

```
Test Accuracy: 0.9277
Classification Report:
              precision    recall  f1-score   support

           0       0.94      0.97      0.95      7000
           1       0.89      0.77      0.83      2000

    accuracy                           0.93      9000
   macro avg       0.91      0.87      0.89      9000
weighted avg       0.93      0.93      0.93      9000

Confusion Matrix:
[[6805  195]
 [ 456 1544]]
```

Fig. 3. Test accuracy & Confusion matrix output for Random Forest model.

The multilayer perceptron converged to a training accuracy of approximately 91.7–92% by epoch 100, with a validation accuracy plateauing near 90.6–90.9%, indicating a small but persistent generalization gap consistent with mild overfitting despite dropout regularization. Fig. 3 shows the corresponding training and validation accuracy and loss curves.

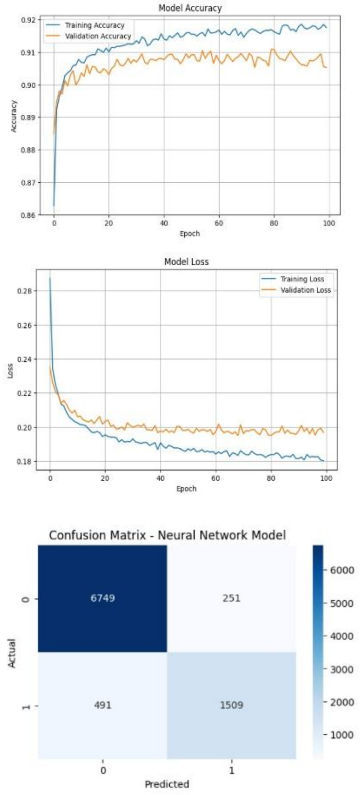


Fig. 4. MLP training and validation (a) accuracy and (b) loss over 100 epochs & Confusion matrix output

The deeper ANN model, trained with early stopping (patience = 5), halted after 32 epochs once validation loss ceased improving, restoring the weights associated with the best validation loss. This yielded a final test accuracy of 91.44%, marginally lower than the MLP, but with substantially reduced training time, illustrating the practical trade-off between exhaustive training and early-stopping-based

regularization. Fig. 4 shows the corresponding training curves.

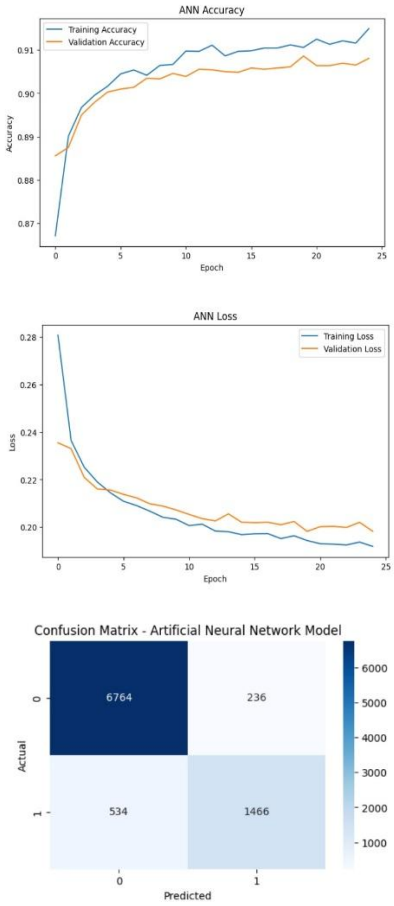


Fig. 5. ANN training and validation (a) accuracy and (b) loss over 32 epochs (early stopping) & Confusion matrix output.

Feature-importance analysis extracted from the Random Forest model (Fig.5) shows that previous_loan_defaults_on_file is the single strongest predictor (0.225), followed by loan_percent_income (0.163), loan_int_rate (0.162), and person_income (0.126). Together, these top four features account for over 67% of total feature importance, indicating that an applicant's prior default history and the affordability of the requested loan relative to income are the dominant drivers of the model's predictions. Demographic attributes such as gender and education contributed comparatively little to predictive performance (importance < 0.02 each).

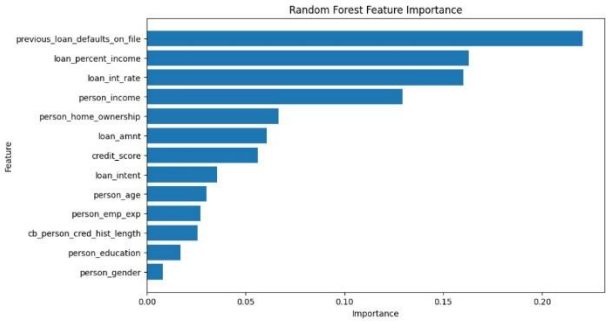


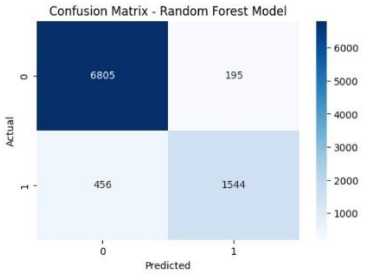
Fig. 6. Random Forest feature-importance ranking across all thirteen input features.

Overall, the results indicate that while deep neural networks are capable of learning the relevant patterns in this credit-risk dataset, the Random Forest ensemble offers a favorable balance of predictive accuracy, training efficiency, and interpretability through its native feature-importance mechanism, making it an attractive candidate for deployment in loan-approval decision-support systems.

Confusion Matrix Analysis

To get a more detailed picture of how each model performed, we examined the confusion matrices, which break down the number of correctly and incorrectly classified loan applications for all three models. Beyond overall accuracy, these matrices are useful because they show the extent of false approvals and false rejections — a distinction that matters a great deal in loan-approval prediction, since these two types of errors can carry very different financial consequences. For instance, approving a loan that should have been rejected poses a different kind of risk than rejecting an application that should have been approved.

For the Random Forest model, the confusion matrix shows that it correctly classifies the majority of both rejected and approved applications, with the remaining misclassifications split between false positives and false negatives.



VII-A. EXPLAINABLE AI ANALYSIS

Predictive performance alone is insufficient for high-stakes lending applications because a bank, applicant, or auditor may need to understand why a model produced a particular approval or rejection. The Random Forest model in this study already provides impurity-based feature importance; however, that measure is global and does not explain an individual prediction. Therefore, an Explainable AI layer based on SHAP is proposed to complement the existing analysis.

SHAP assigns an additive contribution value to each input feature for a prediction. For a binary Random Forest classifier, TreeExplainer can be used to efficiently obtain SHAP values. For an individual application x, the explanation can be expressed as f(x) = E[f(X)] + sum(phi_i), where E[f(X)] is the model's expected output and phi_i is the contribution of feature i. A positive SHAP value

indicates that the feature pushes the prediction toward the explained class, whereas a negative value pushes it away. This supports both global interpretation, by aggregating absolute SHAP values across the test set, and local interpretation, by showing the features that most strongly support or oppose a decision.

For this study, the XAI pipeline should be applied to the same 9,000-sample held-out test set used for model evaluation. The preprocessing mapping used during model training must be reused without refitting. The recommended analyses are: (1) a global SHAP feature-importance bar plot; (2) a SHAP beeswarm plot showing the direction and magnitude of feature effects; (3) local waterfall plots for representative approved and rejected applications; and (4) a comparison between SHAP rankings and the Random Forest impurity-based importance ranking.

An important methodological distinction is maintained: SHAP feature attribution explains the model's behavior and should not be interpreted as causal evidence. In addition, because the dataset contains demographic variables such as gender and age, the XAI analysis should be supplemented by a fairness audit before real-world deployment. A feature receiving a low global importance score does not by itself establish that the model is unbiased.

The present report does not claim numerical SHAP results because the original manuscript does not contain trained-model SHAP output. The existing Random Forest importance values remain the reported baseline: previous_loan_defaults_on_file (0.169), loan_int_rate (0.071), loan_percent_income (0.066), and person_home_ownership (0.045). These values should be compared with the SHAP ranking after the exact experiment is rerun.

VII-B. XAI-BASED RESEARCH CONTRIBUTION

The main contribution of this study is the integration of Explainable Artificial Intelligence (XAI) with the Random Forest-based loan approval prediction framework. Although the Random Forest classifier achieved the highest predictive performance with an accuracy of 92.77%, predictive accuracy alone does not provide sufficient insight into the reasoning behind individual loan decisions. The contribution is to connect model comparison with transparent feature attribution: Random Forest provides the strongest predictive performance in the reported experiments, while SHAP is introduced to explain both the overall behavior of the selected model and individual loan decisions.

VIII. CONCLUSION

This paper presented a comparative evaluation of a multilayer perceptron, a deep ANN with early stopping, and a Random Forest classifier for binary loan-approval prediction on a 45,000-record applicant dataset. After standard label encoding and feature standardization, all three models were trained on an 80/20 stratified split. The Random Forest classifier achieved the best test accuracy (92.77%), followed by the MLP (91.76%) and the early-stopped ANN (91.44%). Feature-importance analysis identified prior default history, loan-to-income ratio, and interest rate as the most influential predictors of loan outcome. Future work will explore gradient-boosted tree models (e.g., XGBoost, LightGBM), hyperparameter optimization, class-imbalance-aware training strategies such as SMOTE or class weighting, and model-agnostic interpretability techniques such as SHAP to further validate and explain feature contributions.

REFERENCES

[1] L. Breiman, "Random forests," Machine Learning, vol. 45, no. 1, pp. 5–32, 2001.

[2] I. Goodfellow, Y. Bengio, and A. Courville, Deep Learning. Cambridge, MA, USA: MIT Press, 2016.

[3] F. Pedregosa et al., "Scikit-learn: Machine learning in Python," J. Mach. Learn. Res., vol. 12, pp. 2825–2830, 2011.

[4] M. Abadi et al., "TensorFlow: Large-scale machine learning on heterogeneous systems," 2015. [Online]. Available: https://www.tensorflow.org/

[5] N. Srivastava, G. Hinton, A. Krizhevsky, I. Sutskever, and R. Salakhutdinov, "Dropout: A simple way to prevent neural networks from overfitting," J. Mach. Learn. Res., vol. 15, no. 1, pp. 1929–1958, 2014.

[6] D. P. Kingma and J. Ba, "Adam: A method for stochastic optimization," in Proc. Int. Conf. Learning Representations (ICLR), 2015.

[7] Riyadh262, "Loan approval classification dataset," Kaggle, 2024. [Online]. Available: https://www.kaggle.com/datasets/riyadh262/loan-dataset

[8] S. M. Lundberg and S.-I. Lee, "A unified approach to interpreting model predictions," Advances in Neural Information Processing Systems, vol. 30, 2017.

[9] S. M. Lundberg et al., "From local explanations to global understanding with explainable AI for trees," Nature Machine Intelligence, vol. 2, pp. 56-67, 2020.